

Let $\operatorname{Re} z_n > 0$ for all n

$\prod_{n=1}^{\infty} z_n$ converges absolutely and is non-zero

$\Leftrightarrow \sum_{n=1}^{\infty} \log z_n$ converges absolutely

$\Leftrightarrow \sum_{n=1}^{\infty} z_n - 1$ converges absolutely

~~~~~

Let  $X$  be a set,  $f_n : X \rightarrow \mathbb{C}$ ,  $f : X \rightarrow \mathbb{C}$ . Let

$f_n \xrightarrow{\text{unif}} f$  on  $X$ . Suppose there exists  $a$  such that

$\operatorname{Re} f(x) \leq a$  for all  $x \in X$ . Then,  $e^{f_n} \xrightarrow{\text{unif}} e^f$  on  $X$ .

~~~~~

Let X be a compact metric space, $g_n : X \rightarrow \mathbb{C}$.

Suppose $\sum_{n=1}^{\infty} (g_n(x) - 1)$ converges uniformly and absolutely on X .

Then, $f(x) = \prod_{n=1}^{\infty} g_n(x)$ converges uniformly and

absolutely on X . Further, there exists n_0 such that $f(x) \neq 0$

$\Leftrightarrow g_n(x) \neq 0$ for some $n \leq n_0$

~~~~~

Let  $G$  be a region,  $\{f_n\} \subset \mathcal{A}(G)$  such that no  $f_n \equiv 0$ .

Suppose  $\sum_{n=1}^{\infty} (f_n(z) - 1)$  converges in  $\mathcal{A}(G)$ . Then,

$f(z) = \prod_{n=1}^{\infty} f_n(z)$  converges in  $\mathcal{A}(G)$ . Further,

$f(z) = 0$  implies only finitely many  $f_n(z) = 0$ .

Let  $\operatorname{Re} z_n > -1$  for all  $n$

$\prod_{n=1}^{\infty} (1 + z_n)$  converges absolutely and is non-zero

$\Leftrightarrow \sum_{n=1}^{\infty} \log(1 + z_n)$  converges absolutely

$\Leftrightarrow \sum_{n=1}^{\infty} z_n$  converges absolutely

~~~~~

Let X be a set, $f_n : X \rightarrow \mathbb{C}$, $f : X \rightarrow \mathbb{C}$. Let

$f_n \xrightarrow{\text{unif}} f$ on X . Suppose there exists a such that

$\operatorname{Re} f(x) \leq a$ for all $x \in X$. Then, $e^{f_n} \xrightarrow{\text{unif}} e^f$ on X .

~~~~~

Let  $X$  be a compact metric space,  $g_n : X \rightarrow \mathbb{C}$ .

Suppose  $\sum_{n=1}^{\infty} g_n(x)$  converges uniformly and absolutely on  $X$ .

Then,  $f(x) = \prod_{n=1}^{\infty} (1 + g_n(x))$  converges uniformly and

absolutely on  $X$ . Further, there exists  $n_0$  such that  $f(x) \neq 0$

$\Leftrightarrow 1 + g_n(x) \neq 0$  for some  $n \leq n_0$

~~~~~

Let G be a region, $\{f_n\} \subset \mathcal{A}(G)$ such that no $f_n \equiv -1$.

Suppose $\sum_{n=1}^{\infty} f_n(z)$ converges in $\mathcal{A}(G)$. Then,

$f(z) = \prod_{n=1}^{\infty} (1 + f_n(z))$ converges in $\mathcal{A}(G)$. Further,

$f(z) = 0$ implies only finitely many $1 + f_n(z) = 0$.